

Program:

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import pandas as pd
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import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
wine=load_wine()
data=pd.DataFrame(data=wine.data,columns=wine.feature_names)
data['Target']=wine.target
x=data.drop('Target',axis=1)
y=data['Target']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)
model=KNeighborsClassifier(n_neighbors=5)
model.fit(x_train,y_train)
y_pred=model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("accuracy:",accuracy)
conf_matrix = confusion_matrix(y_test,y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='summer',
xticklabels=wine.target_names, yticklabels=wine.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```

Output:

